Analysis of Strongest-Users-Only Detectors for Randomly Spread CDMA[1]

Shlomo Shamai (Shitz) [2] Benjamin M. Zaidel [3] Sergio Verdú [4]

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[2]The Department of Electrical Engineering, Technion - IIT, Haifa 32000, Israel. E-mail: sshlomo@ee.technion.ac.il

[3]The Department of Electrical Engineering, Technion - IIT, Haifa 32000, Israel. E-mail: bennyz@inter.net.il

[4]The Department of Electrical Engineering, Princeton University, Princeton, NJ 08544, USA. E-mail: verdu@ee.princeton.edu

Abstract

A practically oriented transmission and decoding strategy is analyzed for a single-cell, randomly spread, direct sequence code division multiple access (DS-CDMA) system and slow flat fading. It is assumed that all users employ equal rates and transmit powers. The receiver ranks the users according to their received powers, and decodes only a subset of strongest users, equal in number to the largest integer for which decoding is successful. The discussion is confined to asymptotic analysis, where both the number of users and the processing gain go to infinity, while their ratio goes to some finite constant. Three multiuser detection strategies are considered: the matched-filter detector, the linear minimum mean squared error (MMSE) detector, and the "optimum detector" under the above strongest-users decoding scheme. The total capacities under an outage constraint of these detection strategies, derived as functions of the fraction of users that *cannot* be reliably decoded (equivalent to the outage probability), are analyzed and compared. With a proper system design, the outage constrained capacity of the "optimum detector" is shown to approach the optimum spectral efficiency asymptotically with $\frac{E_b}{N_0}$.

1 Introduction

Aiming at the *maximum* achievable throughput, or "spectral efficiency", the ultimate performance of single-cell, randomly spread, direct sequence code division multiple access (DS-CDMA) systems has been thoroughly analyzed in [1] and [2]. Focusing on the limiting scenario, where both the number of users and the processing gain go to infinity, while their ratio goes to some *finite* constant, deterministic system performance measures of interest were obtained, using relevant random matrix theory results (e.g., [3] – [4]). In particular, assuming equal received powers and no fading, expressions are presented in [1] for the spectral efficiency of the optimum detector, the matched-filter detector, the decorrelator, and the linear minimum mean squared error (MMSE) detector. The results are extended in [2] to examine the impact of frequency-flat fading on these receivers. Analogous results in a simple multi-cell scenario can be found in [5] – [7].

Tacit in the spectral efficiency analysis [2] is the assumption that all active users are (reliably) decoded *regardless of their received powers.* In slow-fading channels, where the channel fading gains are assumed to be constant within the codeword duration, the above assumption actually implies that the users adjust their rates (and possibly their powers) as a function of the channel fading level they experience, via delayless feedback from the receiver. Unfortunately, in practice, such feedback and ideal tuning of the users' transmissions cannot always be accomplished, and motivated by *practical* considerations weanalyze in this report an alternativeapproac h. Accordingly, excluding any form of feedback from the receiving cell-site, but an ACK/NACK type, it is assumed that all users transmit at *equal rates* and *equal powers* regardless of the individual fade levels (note that in [2] *rate* adjustment feedback is still tacitly assumed in the equal transmit power setup). In such case, due to the presence of fading, the receiver at the cell-site can no longer guarantee reliable decoding of all active users. The notion of *"outage probability"* arises then as an appropriate system performance specifying parameter, defined as the probability that a randomly selected user will not be received with strong enough power to be successfully decoded (a scheduling protocol can be combined in this setup so that undecoded users are eventually decoded reliably). It should be noted at this pointthat this comes in contrast to the setup of (relatively) *fast*-fading channels where the per-user fading process is stationary and ergodic, in which case *all* users can be reliably decoded

at the receiver while employing (appropriately chosen) fixed and equal rates (see [8] and references therein). Methods for obtaining the outage probability of linear CDMA receivers as a function of the rate, $\frac{E_b}{N_0}$, and the fading distribution, are shown in [2] and [9].

In this report, independent (over users) identically distributed flat-fading channels are assumed, as in [2], and system performance is analyzed while employing *optimally* coded *randomly* spread DS-CDMA with multiuser detection. As in [1] – [2], the limiting scenario is considered in which the number of active users K, and the spreading factor N (processing gain), both go to infinity, while $\frac{K}{N} \rightarrow \beta < \infty$. The factor β is commonly referred to as the "system load". Applying the aboveeq ual transmission rate and power strategy, it is assumed that the cell-site receiver ranks all active users by their received powers, and then only decodes the transmissions of a *subset* of these users, comprising the users with strongest received powers (hence the name "strongest-users decoding"). The number of decoded users is chosen as the largest integer for which decoding is successful (this number depends on the realizations of both the fade levels and the additive background noise). The above strategy was first considered in [10] for the case in which no spreading is employed (and all bandwidth is available for coding). The underlying assumption of our analysis is that the system designer sets the transmission rate of all users in order to achieve a target fraction of *undecodable* users (FUU), which is completely equivalent to the notion of outage probability. Since different users experience independent fades, as the number of users grows, the percentage of undecodable users converges to a deterministic constant. Specifically, denoting by J the number of (reliably) *decoded* users, it is assumed that at the limiting scenario $\frac{J}{K} \rightarrow 1 - \mathcal{Q}$, where $\mathcal{Q} \in [0, 1)$ denotes the limiting system FUU. Optimization of the overall throughput in this regime, by choosing the optimum $\mathcal{Q}$, is also considered. The rationale behind such optimization is the tradeoff between the (fixed and equal) rate to be employed by each individual user, and the FUU. As shown in the following, the above tradeoff does not lead to trivial results (i.e., $\mathcal{Q} \rightarrow 0, 1$) for the optimum $\mathcal{Q}$.

Focusing on the single-cell setup, three types of multiuser detection strategies are considered:

1). The *"conventional" matched-filter detector* that treats *all* interference as additive white Gaussian noise;

2). The *linear MMSE detector* that knows the signatures (spreading sequences) of all interfering users and mitigates their interference by means of a linear MMSE filter;

3). A detector that *optimally* decodes the transmissions of the maximum decodable subset of users, while taking into account the structure of the multiuser interference generated by all remaining undecodable users [11] (assuming their signatures are known at the receiver). This detector is referred to henceforth as the "optimum detector", where it is to be understood that optimality here is under the constraints of the strongest-users decoding scheme.

These multiuser detection strategies are analyzed and compared in terms of their total achievable rate sum over all decodable users, under the strongest-users decoding scheme, to be referred to henceforth as the *"outage constrained capacity"*. The outage constrained capacity of the two linear detectors is derived following [2]. The expression for the outage constrained capacity of the "optimum detector" has been derived in [11]. In this context, it is assumed that the system designer may in general modify the target FUU as a function of the signal-to-noise ratio (SNR), or $\frac{E_b}{N_0}$, to optimize the outage constrained capacity. It is noted however that in some scenarios, practical considerations such as the receiver's complexity, or its available resources (translating to a maximum number of simultaneously decodable users), may restrict the minimum FUU. System performance can be further optimized by modifying the system load β. Analogous results in the case in which no spreading is employed are presented in [10], considering the matched-filter equivalent detector, that treats all interference as additive noise, the "optimum detector", and a detector that performs successive interference cancellation, using the matched-filter equivalent detector at each stage of the cancellation process.

The structure of this report is as follows. Section 2 describes the system model. Section 3 presents the expressions for the outage constrained capacities of each of the three multiuser detection strategies. Section 4 includes an analysis and some numerical results. Finally, Section 5 terminates this report with some concluding remarks.

2 System Model

Using the standard discrete time equivalent channel representation, the signal vector received at the cell-site, at the discrete time related to the transmission of the ith symbol, is given by

$$\boldsymbol{y}_i = \boldsymbol{S}_i \boldsymbol{H}_i \boldsymbol{x}_i + \boldsymbol{n}_i \ . \tag{2-1}$$

The vector $\boldsymbol{x}_i = [x_{1,i}, \ldots, x_{K,i}]^T$ in (2-1) comprises the K code symbols transmitted by the users at the ith discrete time. These code symbols are assumed to be i.i.d. circularly symmetric complex Gaussian random variables (which conforms with the capacity achieving statistics), with $E\{x_{k,j}\} = 0$ and $E\{|x_{k,j}|^2\} = \bar{P}\ \forall k, j$, where $\bar{P}$ is the equal transmit power of all users. This model is justified by assuming that the codebooks of all users are chosen randomly, governed by an underlying i.i.d. Gaussian distribution per symbol, and *independently* for each message transmission [12]. The columns of the $N \times K$ matrix $\boldsymbol{S}_i$ are the spreading sequences (signatures) of the users. The entries of $\boldsymbol{S}_i$ are treated as i.i.d. zero-mean random variables, with variance $1/N$. The vector $\boldsymbol{n}_i$ represents a zero-mean white circularly symmetric complex Gaussian noise vector, with $E\{\boldsymbol{n}_i \boldsymbol{n}_i^\dagger\} = \boldsymbol{I}$, $\forall i$. Without loss of generality all received powers are thus normalized with respect to the noise spectral level, and represent in fact the *SNRs* at the input to the multiuser detectors. Finally, $\boldsymbol{H}_i \triangleq \text{diag}\left(h_{1,i}, \ldots, h_{K,i}\right)$, where $\{h_{k,i}\}_{k=1}^K$ designate the assumed i.i.d. zero-mean channel fading gains associated with the signals of the different users, at the ith discrete time. In this setup, as the system size becomes large ($N, K \to \infty$, $\frac{K}{N} \to \beta < \infty$), the empirical distribution of the channel fading (power) levels, $\nu_{k,i} \triangleq |h_{k,i}|^2$, converges almost surely (a.s.) to a distribution $\mathcal{F}_\nu$. The fading levels are assumed throughout to be normalized so that $E_{\mathcal{F}_\nu}\{\nu\} = 1$. It is noted that the analysis to follow applies for a general fading distribution, while assuming that its density $\frac{\partial \mathcal{F}_\nu(\nu)}{\partial \nu}$ exists and is continuous on its supports [10]. Rayleigh fading is assumed whenever explicit results are obtained.

3 Outage Constrained Capacity

3.1 Preliminaries

The outage constrained capacity of a linear detector for a given FUU, is most conveniently expressed in terms of its *multiuser efficiency* [13], defined as the ratio between the detector's output signal-to-interference-plus-noise ratio (SINR) and the SNR (note that multiuser efficiency may depend on the presence of fading [2]). The derivation of the outage constrained capacity of the "optimum detector" [11] (see Theorem 3.1 below) is more involved, and the details are reviewed in Appendix B. The reader is also referred to Appendix A for a short summary of some basic order statistics results that are used in the derivations to follow.

For the matched-filter and linear MMSE detectors, the multiuser efficiency is in fact equal for all users and independent of the instantaneous fading realization experienced by any particular user at any time instant. Denoting the multiuser efficiency by η, and following central limit results showing that the interference at the output of each of the two linear detectors is well approximated by a Gaussian noise (see [1], [5] and references therein for justification of this Gaussian approximation), the kth strongest user is decodable if the (equal) transmission rate per user (bits/sec/Hz) satisfies:

$$\tilde{R} \leq \frac{1}{N} \log\left(1 + \nu_{i_k} \eta \bar{P}\right) \ , \tag{3-1}$$

where $i_k$denote s the index of the kth strongest user at the receiver (i.e., $\nu_{i_1} \geq \nu_{i_2} \geq \cdots \geq \nu_{i_K}$). Hence, the outage constrained capacity for a given FUU of $\mathcal{Q}$ equals

$$\begin{aligned} \tilde{R}_T(\mathcal{Q}) &= \lim_{\substack{K,N,J\to\infty, \\ \frac{K}{N}\to\beta, \frac{J}{K}\to 1-\mathcal{Q}}} \frac{1}{N} J \log\left(1 + \nu_{i_J} \eta \bar{P}\right) \\ &= \beta(1-\mathcal{Q}) \log\left(1 + \mathcal{F}_{\nu}^{-1}(\mathcal{Q}) \eta \bar{P}\right) \ . \end{aligned} \tag{3-2}$$

It is noted that when different systems are to becom pared (with possibly different spreading gains and data rates), it is useful to express the outage constrained capacity in

terms of the system average $\frac{E_b}{N_0}$, which is done through the relation

$$\bar{P} = \frac{1}{\beta}\,\tilde{R}_T\,\frac{E_b}{N_0}\ . \tag{3-3}$$

Substituting (3-3) in (3-2), focusing on the low SNR regime (i.e., $\bar{P} \to 0$, which is equivalent to taking $\tilde{R}_T(\mathcal{Q}) \to 0$), and observing the limiting behavior of the multiuser efficiency (see (3-8) and (3-10) below), the minimum received $\frac{E_b}{N_0}$ that enables reliable communications is found to be

$$\frac{E_b}{N_0}_{\min} = \frac{\ln 2}{(1-\mathcal{Q})\mathcal{F}_\nu^{-1}(\mathcal{Q})}\ . \tag{3-4}$$

The penalty in $\frac{E_b}{N_0}_{\min}$ induced by the strongest-users decoding scheme is hence a factor of $[(1-\mathcal{Q})\mathcal{F}_\nu^{-1}(\mathcal{Q})]^{-1}$, as compared to $\frac{E_b}{N_0}_{\min} = \ln 2 \approx -1.59\text{dB}$, the corresponding value when *all* users are decoded [2]. Although less obvious, the same result can be verified for the "optimum detector" as well. The above penalty is explained by the power "wasted" when users fail to be decoded. The reader is referred to Appendix C for more details on the derivation of (3-4).

For the particular case of Rayleigh fading

$$\mathcal{F}_\nu(y) = 1 - e^{-y},\ \mathcal{F}_\nu^{-1}(x) = -\ln(1-x),\ d\mathcal{F}_\nu(y) = e^{-y}\,dy\ , \tag{3-5}$$

and the outage constrained capacity of linear detectors is given by

$$\tilde{R}_T(\mathcal{Q}) = \beta(1-\mathcal{Q})\log\left(1 - \ln(1-\mathcal{Q})\eta\bar{P}\right)\ . \tag{3-6}$$

Applying (3-4), it follows that for *all* three multiuser detection strategies

$$\frac{E_b}{N_0}_{\min} = -\frac{\ln 2}{(1-\mathcal{Q})\ln(1-\mathcal{Q})}\ , \tag{3-7}$$

the minimum of which is $\frac{E_b}{N_0}_{\min} = e\ln 2 \approx 2.75\text{dB}$,ac hieved for $\mathcal{Q} = 1-1/e$ (being hence the *optimum FUU* in the low SNR regime, for Rayleigh fading channels). As can be observed, the strongest-users decoding scheme induces a *severe* penalty of $10\log_{10} e \approx$ 4.34dB in Rayleigh fading channels. The same degradation has been identified in the broadcast strategy of communicating over (slowly) fading channels with no rate feedback to the

transmitter [14]. According to the latter strategy, the transmitter views the fading channel as a degraded Gaussian broadcast channel with a continuum of receivers each experiencing a different signal to noise ratio according to the possible values of the fading level (assumed continuous as well). The transmitter is then envisaged as transmitting in parallel infinitely many information streams with rates corresponding to the possible fade levels. Each of the above streams can be reliably decoded by receivers corresponding to equal or lower fade level. The analogy to the setting considered in this report is straightforwardly observed.

The following subsections present the outage constrained capacity results of each of the three multiuser detection strategies, under the strongest-users decoding scheme. The notation $(\cdot)_{\text{mf}}$, $(\cdot)_{\text{ms}}$ and $(\cdot)_{\text{opt}}$ is used to designate entries related to the matched-filter detector, the linear MMSE detector, and the "optimum detector", respectively. For simplicity of notation, equations are expressed in terms of the transmit power $\bar{P}$ (which is in fact the SNR, following the normalization with respect to the noise spectral level).

3.2 The Matched-Filter Detector

As shown in [1], [2] and [15], the multiuser efficiency of the matched-filter detector, in the limiting scenario considered here, converges in probability to

$$\eta_{\text{mf}} = \frac{1}{1+\beta\bar{P}} \,. \tag{3-8}$$

The outage constrained capacity of the matched-filter detector for a target FUU of $\mathcal{Q}$ is hence given by:

$$\begin{aligned} \tilde{R}_{T\,\text{mf}}(\mathcal{Q}) &= \beta(1-\mathcal{Q})\log\left(1+\mathcal{F}_{\nu}^{-1}(\mathcal{Q})\frac{\bar{P}}{1+\beta\bar{P}}\right) \\ &\underset{\text{Rayleigh}}{=} \beta(1-\mathcal{Q})\log\left(1-\ln(1-\mathcal{Q})\frac{\bar{P}}{1+\beta\bar{P}}\right) \,. \end{aligned} \tag{3-9}$$

3.3 The Linear MMSE detector

The multiuser efficiency of the linear MMSE detector, [15], [2], converges a.s. as $K, N \to \infty$, $\frac{K}{N} \to \beta < \infty$, to a non-random limit given by the unique positive solution to the following

implicit equation

$$\begin{aligned} 1 &= \eta_{\text{ms}} + \beta E_{\mathcal{F}_\nu}\left\{\frac{\bar{P}\nu\eta_{\text{ms}}}{1+\bar{P}\nu\eta_{\text{ms}}}\right\} \\ &\underset{\text{Rayleigh}}{=} \eta_{\text{ms}} + \beta\left[1 - \frac{1}{\bar{P}\eta_{\text{ms}}}\, e^{\frac{1}{\bar{P}\eta_{\text{ms}}}}\, \mathcal{E}_1\!\left(\frac{1}{\bar{P}\eta_{\text{ms}}}\right)\right], \end{aligned} \tag{3-10}$$

where $\mathcal{E}_1(x) \triangleq \int_x^\infty \frac{e^{-t}}{t}\, dt$, $t \geq 0$, is the exponential integral function. The outage constrained capacity for a target FUU of $\mathcal{Q}$ is evaluated by substituting the result into (3-2) (or (3-6) for Rayleigh fading).

3.4 The "Optimum Detector"

The outage constrained capacity of the "optimum detector" under the strongest-users decoding scheme, for a target FUU of $\mathcal{Q}$, is given by the following theorem.

Theorem 3.1 (Shamai–Verdú [11]) *Let $\eta_{opt}(\delta\,;\bar{P})$ be the unique solution to the equation*

$$1 - \beta(1-\delta) = \eta - \beta\Theta_{opt}(\delta\,;\bar{P}\eta)\ , \tag{3-11}$$

where

$$\begin{aligned} \Theta_{opt}(\omega\,;\varsigma) &\triangleq \int_{\mathcal{F}_\nu^{-1}(0)}^{\mathcal{F}_\nu^{-1}(1-\omega)} \frac{1}{1+\varsigma\phi}\, d\mathcal{F}_\nu(\phi) \\ &\underset{\text{Rayleigh}}{=} \int_0^{-\ln\omega} \frac{1}{1+\varsigma\phi}\, e^{-\phi}\, d\phi \\ &= \frac{1}{\varsigma}\, e^{\frac{1}{\varsigma}} \left[\mathcal{E}_1\!\left(\frac{1}{\varsigma}\right) - \mathcal{E}_1\!\left(\frac{1}{\varsigma} - \ln\omega\right)\right]. \end{aligned} \tag{3-12}$$

Define

$$\begin{aligned} \Gamma_{opt}(w\,;\varsigma) &\triangleq \int_{\mathcal{F}_\nu^{-1}(0)}^{\mathcal{F}_\nu^{-1}(1-\omega)} \log\left(1+\varsigma\phi\right)\, d\mathcal{F}_\nu(\phi) \\ &\underset{\text{Rayleigh}}{=} \int_0^{-\ln\omega} \log\left(1+\varsigma\phi\right) e^{-\phi}\, d\phi \\ &= -\omega\log(1-\varsigma\ln\omega) + e^{\frac{1}{\varsigma}} \left[\mathcal{E}_1\!\left(\frac{1}{\varsigma}\right) - \mathcal{E}_1\!\left(\frac{1}{\varsigma} - \ln\omega\right)\right]\log e \end{aligned} \tag{3-13}$$

and

$$\mathcal{C}_{opt}(\delta\,;\bar{P}) \triangleq \beta\Gamma_{opt}\left(\delta\,;\bar{P}\eta_{opt}(\delta\,;\bar{P})\right) - \log\eta_{opt}(\delta\,;\bar{P}) + \left(\eta_{opt}(\delta\,;\bar{P}) - 1\right)\log e\ . \tag{3-14}$$

Then, the outage constrained capacity of the "optimum detector", for a target FUU of $\mathcal{Q}$, is given by

$$\tilde{R}_{T_{opt}}(\mathcal{Q}) = \inf_{0 \le x < 1} \frac{\mathcal{C}_{opt}\left[(1-\mathcal{Q})(1-x)\,;\bar{P}\right] - \mathcal{C}_{opt}\left[(1-\mathcal{Q})\,;\bar{P}\right]}{x} \,. \tag{3-15}$$

Proof: See Appendix B. ■

4 Analysis and Numerical Results

Comparative outage constrained capacity results in Rayleigh fading channels, for the three multiuser detectors, are presented in Fig. 1 on the following page. The outage constrained capacities of all detectors were evaluated for the (numerically obtained unless otherwise stated) optimum values of both the system load β and $\mathcal{Q}$. Both optimum parameters are in general functions of $\frac{E_b}{N_0}$. For the sake of comparison, the *spectral efficiencies* of the three detectors (as reproduced from [2]) are also included in Fig. 1. Fig. 2 on page 11 shows the corresponding optimum FUU as a function of $\frac{E_b}{N_0}$.

Careful examination of (3-9) shows that the outage constrained capacity of the matched-filter detector monotonically increases with the system load β. The limiting outage constrained capacity for Rayleigh fading is given by

$$\lim_{\beta\to\infty} \tilde{R}_{T_{\mathrm{mf}}}(\mathcal{Q}) = -(1-\mathcal{Q})\log(1-\mathcal{Q}) - \left(\frac{E_b}{N_0}\right)^{-1} \,, \tag{4-1}$$

which coincides with the corresponding result in the non-spreading regime [10] (using the equivalence of β and K in the latter case). The optimum FUU for *all* $\frac{E_b}{N_0}$ values, in Rayleigh fading channels, is hence $\mathcal{Q} = 1 - 1/e$ for which

$$\lim_{\beta\to\infty} \tilde{R}_{T_{\mathrm{mf}}}(\mathcal{Q})|_{\mathcal{Q}=1-1/e} = \frac{\log e}{e} - \left(\frac{E_b}{N_0}\right)^{-1} \,. \tag{4-2}$$

The detector is obviously interference limited, and the outage constrained capacity reaches the limit of $\frac{\log e}{e} = 0.5307$ bits/sec/Hz as $\frac{E_b}{N_0} \to \infty$, as can be seen in Fig. 1 (the corresponding *spectral efficiency* of the detector reaches the limit of $\log e = 1.4427$).

In contrast to the matched-filter detector, the linear MMSE detector is not interference

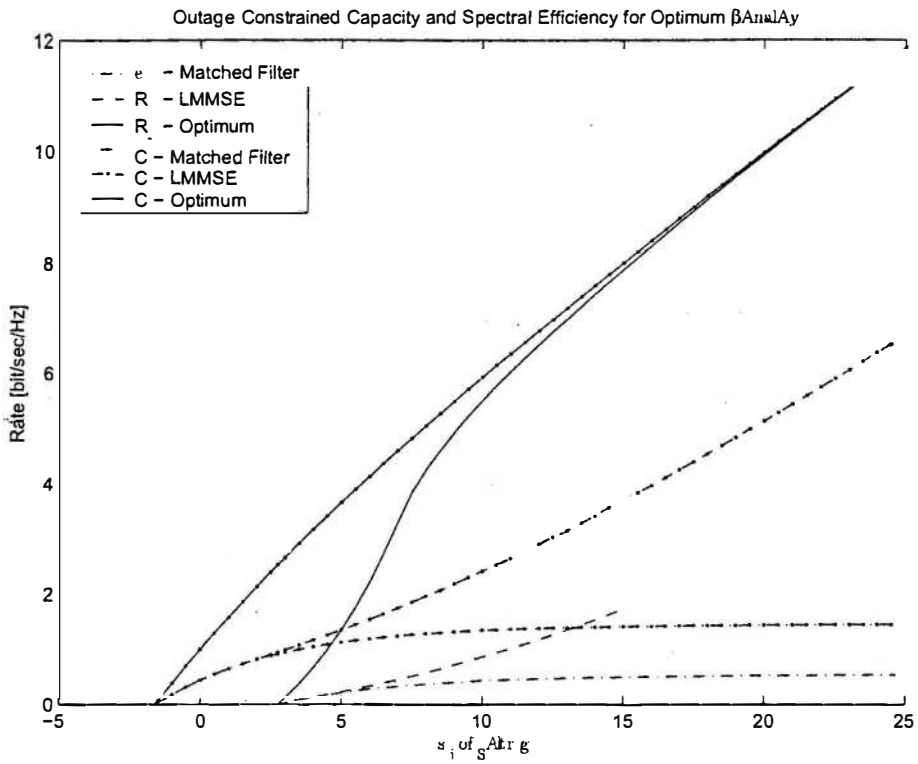


Figure 1: Spectral efficiency and outage constrained capacity of the multiuser detectors, for optimum choice of system load β and FUU (Rayleigh fading).

limited, and its *optimized* outage constrained capacity grows without bound with $\frac{E_b}{N_0}$. For low $\frac{E_b}{N_0}$ values, it is optimum to take $\beta \to \infty$, in which case the outage constrained capacity of the linear MMSE detector coincides with that of the matched-filter detector. However, as $\frac{E_b}{N_0}$ is increased, the optimum value of β starts to decrease beyond a critical value, eventually becoming lower than unity. A similar behavior is also observed while examining the spectral efficiency of the detector, as discussed in [2]. The critical $\frac{E_b}{N_0}$ value above which the optimum β in terms of *spectral efficiency* takes on finite values is explicitly identified in [2] and given by the following Lemma.

Lemma 4.1 (Shamai–Verdú, [2] Theorem III.4) *Suppose that* $E_{\mathcal{F}_\nu}\{\nu^2\} < \infty$. *Then, for the linear MMSE detector, the optimum value in terms of spectral efficiency of the*

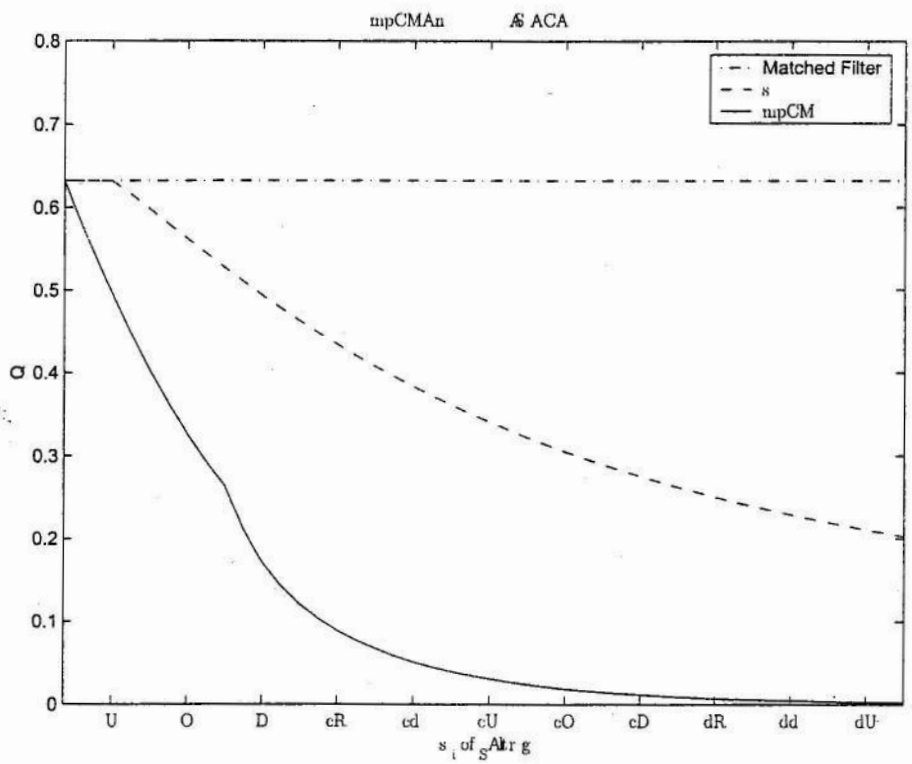


Figure 2: Optimum values of $\mathcal{Q}$ for the three multiuser detectors (Rayleigh fading).

system load β is finite if and only if

$$\frac{E_b}{N_0} > 2\ln 2 = 1.41855 dB\ .$$

Using similar arguments to those used for proving the above Lemma, an analogous result for the outage constrained capacity can beobtained. The result is summarized by the following theorem.

Theorem 4.2 *Suppose that $E_{\mathcal{F}_\nu}\{\nu^2\} < \infty$. Then, for the linear MMSE detector, the optimum value in terms of the outage constrained capacity of the system load β is finite if and only if*

$$\frac{E_b}{N_0} > \frac{2E_{\mathcal{F}_\nu}\{\nu^2\}}{2E_{\mathcal{F}_\nu}\{\nu^2\} - \mathcal{F}_\nu^{-1}(\mathcal{Q})} \frac{E_b}{N_{0\,min}} \underset{\text{Rayleigh}}{=} \frac{4}{4 + \ln(1-\mathcal{Q})} \frac{E_b}{N_{0\,min}}\ , \tag{4-3}$$

where $\frac{E_b}{N_0}_{min}$ *is given by* (3-4).

Proof: See Appendix D. ■

The optimum FUU monotonically decreases with $\frac{E_b}{N_0}$, beyond the above critical $\frac{E_b}{N_0}$ value (below which it equals $1-1/e$ for Rayleigh fading channels), as observed in Fig. 2 on the preceding page. The critical $\frac{E_b}{N_0}$ corresponds to the point beyond which the optimum value for the system load β starts to decrease from infinity and the linear MMSE detector outperforms the matched-filter detector. The value for $\mathcal{Q} = 1-1/e$ (according to Theorem 4.2) equals $\frac{4}{3}e\ln 2 \approx 4dB$.

The "optimum detector" is also not interference limited, provided that *both* the system load β and the FUU are appropriately chosen. The obtained results show that it is optimum in terms of the outage constrained capacity to take $\beta \to \infty$ for all $\frac{E_b}{N_0}$ values. Evaluating the limit of (3-15) as $\beta \to \infty$, it is observed that the outage constrained capacity of the "optimum detector", for Rayleigh fading channels, satisfies the following equation:

$$\lim_{\beta\to\infty} \tilde{R}_{T_{\text{opt}}}(\mathcal{Q}) = \inf_{x\in[0,1)} \frac{1}{x}\log\left[1+\frac{\beta\bar{P}(1-\mathcal{Q})\left[x\left[1-\ln(1-\mathcal{Q})\right]+(1-x)\ln(1-x)\right]}{1+\beta\bar{P}\left[\mathcal{Q}+(1-\mathcal{Q})\ln(1-\mathcal{Q})\right]}\right], \tag{4-4}$$

which coincides with the analogous outage constrained capacity when no spreading is employed [10]. Considering this limiting regime, the outage constrained capacity, while optimized with respect to the FUU, approaches the *optimum spectral efficiency* (coinciding in this case with the *single user* additive white Gaussian noise channel capacity) at the high $\frac{E_b}{N_0}$ region, as observed in Fig. 1. As shown in [10], by letting $\mathcal{Q}$ vanish as $\sqrt{\frac{2\epsilon}{\beta\bar{P}}}$, for some $\epsilon > 0$, the total achievable throughput satisfies

$$\lim_{\beta\bar{P}\to\infty} \tilde{R}_{T_{\text{opt}}}(\mathcal{Q}) = \log\left(1+\frac{\beta\bar{P}}{1+\epsilon}\right). \tag{4-5}$$

Comparing the outage constrained capacity of the "optimum detector" to the optimum spectral efficiency at the *low* $\frac{E_b}{N_0}$ region, the strongest-users decoding scheme is observed to induce a severe penalty in system performance, emphasizing the crucial role of *rate*-adjustment feedback in this regime.

It is interesting to note at this point, that taking $\beta \to \infty$ (in fact if it exceeds a threshold) is *no longer the optimum choice* for the "optimum detector" in the case in

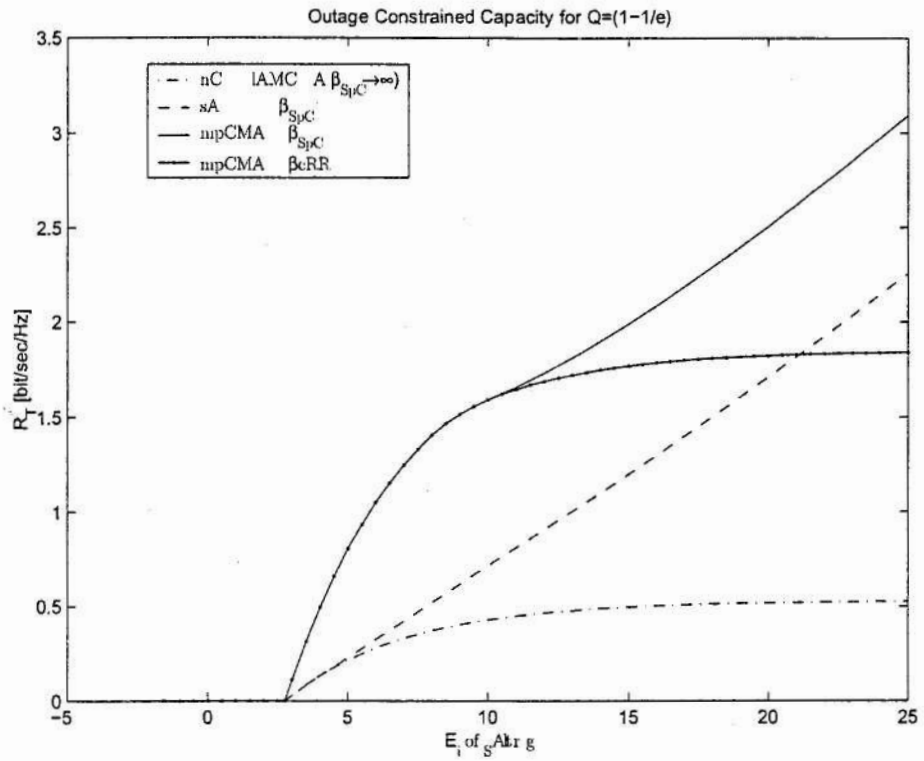


Figure 3: Outage constrained capacity of the multiuser detectors for $\mathcal{Q} = 1-1/e$ (Rayleigh fading).

which, due to some practical implementation considerations, the minimum FUU cannot go below some fixed positive value. As can be observed from (4-4), the "optimum detector" becomes in fact *interference limited* for any *fixed* $\mathcal{Q}$. It can be therefore concluded, that in such case incorporating some level of spreading is *beneficial* in terms of system performance in the high $\frac{E_b}{N_0}$ region, and not all bandwidth should be made available for coding. This somewhat surprising behavior is in sheer contrast to that of the optimum spectral efficiency, for which no spreading is optimum, and it emanates from the inherentsub optimality of the equal-rate equal-transmit-power strongest-users decoding scheme. This phenomena is clearly demonstrated in Fig. 3 where the outage constrained capacities, for Rayleigh fading channels, are plotted for the optimum system load β and with the FUU fixed to $\mathcal{Q} = 1 - 1/e$, the optimum value in the low $\frac{E_b}{N_0}$ regime (and in fact the optimum for all $\frac{E_b}{N_0}$ values for the matched-filter detector). The high β behavior of the "optimum detector" is

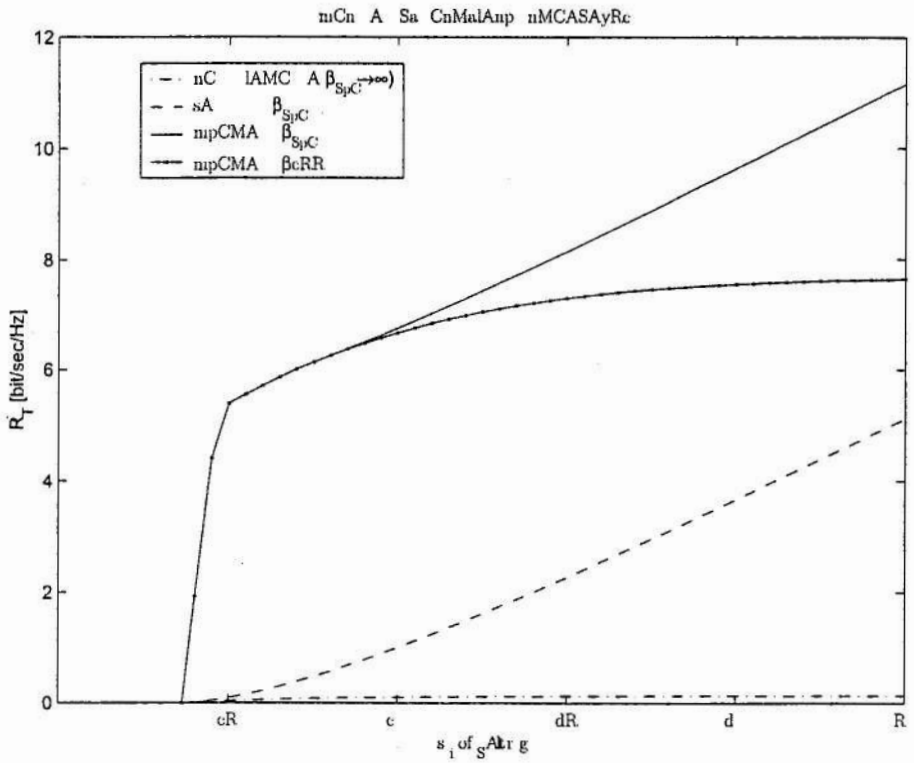


Figure 4: Outage constrained capacity of the multiuser detectors for $\mathcal{Q} = 0.1$ (Rayleigh fading).

demonstrated here by taking $\beta = 100$. The corresponding results for $\mathcal{Q} = 0.1$ are plotted in Fig. 4. As can be observed, the performance of both the linear MMSE detector and the "optimum detector" is considerably enhanced in the latter example (as compared to Fig. 3), at the high $\frac{E_b}{N_0}$ region. In contrast, the performance of the matched-filter detector is severely degraded for all $\frac{E_b}{N_0}$ values as compared to the optimum choice of $\mathcal{Q}$. Fig. 5 on the next page shows the optimum system load β for the "optimum detector" as a function of $\frac{E_b}{N_0}$, for $\mathcal{Q} = 0.1$. At low $\frac{E_b}{N_0}$ values it is optimum to take $\beta \to \infty$ (the maximum value in Fig. 5 was truncated at $\beta = 10$). However, beyond some critical value the optimum system load is observed to be monotonically decreasing with $\frac{E_b}{N_0}$, reaching a value of $\beta \approx 1.78$ at $\frac{E_b}{N_0} = 30$dB.

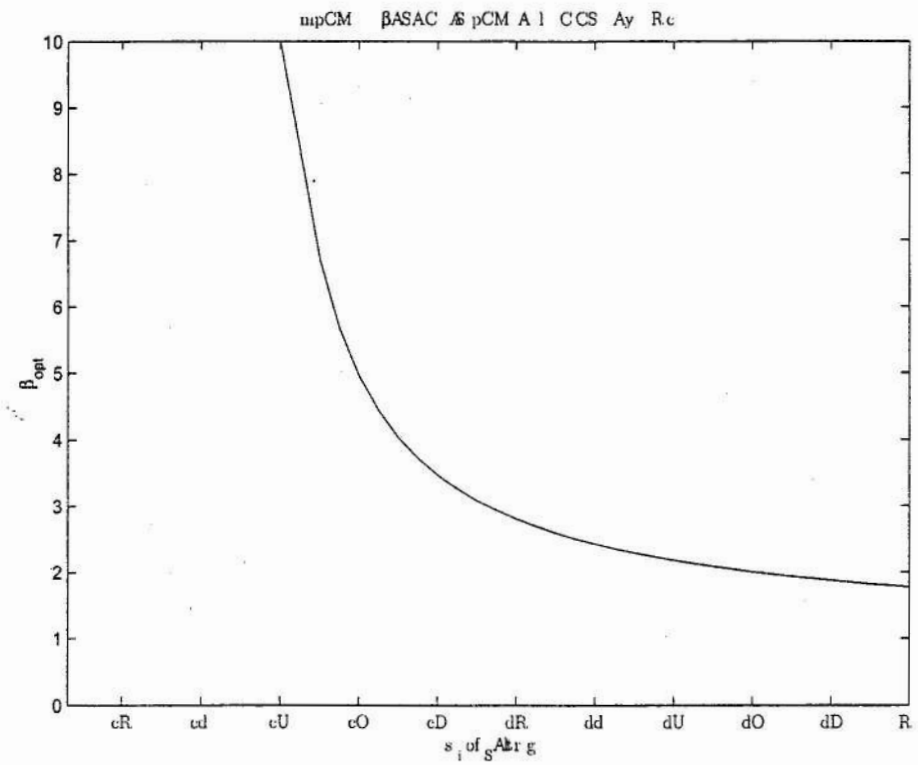


Figure 5: The optimum system load β for the "optimum detector" for $\mathcal{Q} = 0.1$ (Rayleigh fading). The maximum value is truncated at $\beta = 10$.

5 Concluding Remarks

We addressed a practically appealing transmission and decoding strategy in randomly spread DS-CDMA systems. Accordingly, all users are assumed to employ equal and constant transmission powers and rates, the receiver ranks all users according to their *received* powers, and decodes only a subset of strongest users. The FUU is assumed to be a system design parameter.

Considering the outage constrained capacity of the matched-filter detector, the linear MMSE detector, and the "optimum multiuser detector", the strongest-users decoding scheme was shown to induce a severe penalty in the minimum $\frac{E_b}{N_0}$ allowing reliable communications. For Rayleigh fading the penalty is of at least $10 \log_{10} e \approx 4.34$dB. The above minimum penalty is attained by setting the FUU to $\mathcal{Q} = 1 - 1/e$, the optimum choice at

the low SNR regime.

Both linear detectors suffer a significant performance degradation for all $\frac{E_b}{N_0}$ values, as compared to their spectral efficiency (specifying the ultimate performance [2]). In contrast, the outage constrained capacity of the "optimum detector" is shown to asymptotically approach the optimum spectral efficiency in the high $\frac{E_b}{N_0}$ region, when the system load is made large ($\beta \to \infty$), and when the FUU vanishes at an appropriate rate as $\frac{E_b}{N_0} \to \infty$. This asymptotic approach emphasizes the crucial role of *rate-adjustment feedback* at the *low* SNR regime, where the strongest-users decoding scheme induces a *severe degradation* in system performance. An interesting observation is that when the FUU is *fixed*, the "optimum detector" becomes *interference limited* in the large system load region. In such case, the use of some amount of spreading ($\beta < \infty$) turns out beneficial beyond some critical $\frac{E_b}{N_0}$, and the outage constrained capacity of the detector grows without bound with $\frac{E_b}{N_0}$ for an appropriately chosen system load. This behavior is due to the inherent suboptimality of the strongest-users decoding scheme.

Extension of the above results to a simple multi-cell scenario as in [5] – [7] is provided in [16].

A Some Useful Order Statistics Results

Let i_k denote the index of the kth strongest user at the receiver, i.e.,

$$\nu_{i_1} \geq \nu_{i_2} \geq \cdots \geq \nu_{i_K} \ . \tag{A-1}$$

Using $\mathcal{F}_\nu$ to denote the cumulative probability distribution (cdf) of each of the i.i.d. fading levels $\{\nu_m\}$, the distribution of ν_{i_k} is given by (e.g., [13] pp. 388)

$$\mathcal{F}_{\nu_{i_k}}(x) = \int_0^x \frac{K!}{(K-k)!(k-1)!} \left(1 - \mathcal{F}_\nu(z)\right)^{k-1} \mathcal{F}_\nu(z)^{K-k} d\mathcal{F}_\nu(z) \ , \tag{A-2}$$

which for $K \to \infty$ yields [10]

$$\lim_{K\to\infty} \mathcal{F}_{\nu_{i_k}}(x) = \begin{cases} 0 & 1 - \mathcal{F}_\nu(x) > \frac{k-1}{K-1}, \\ 1 & 1 - \mathcal{F}_\nu(x) \leq \frac{k-1}{K-1}. \end{cases} \xrightarrow[K,k\to\infty \frac{k}{K}\to\delta\in(0,1]]{} \begin{cases} 0 & \mathcal{F}_\nu(x) < 1-\delta, \\ 1 & \mathcal{F}_\nu(x) \geq 1-\delta. \end{cases} \tag{A-3}$$

The asymptotic expectation of the kth strongest fading (power) level is given by [11]

$$\begin{aligned} E\{\nu_{i_k}\} &= \int_0^\infty \left(1 - \mathcal{F}_{\nu_{i_k}}(x)\right) dx \\ &\xrightarrow[K,k\to\infty,\, \frac{k}{K}\to\delta\in(0,1]]{} \int_0^\infty 1\{1 - \mathcal{F}_\nu(x) > \delta\}\, dx + o(1) \\ &= \mathcal{F}_\nu^{-1}(1-\delta) + o(1)\ , \end{aligned} \tag{A-4}$$

where $1\{\cdot\}$ designates the indicator function and $\mathcal{F}_\nu^{-1}(\cdot)$ denotes the inverse function of the cdf $\mathcal{F}_\nu(x)$. Finally, for a continuous and differentiable function $g(x)$ it follows that [11]

$$\frac{1}{K}\sum_{k=k_L}^{k_H} g(\nu_{i_k}) \xrightarrow[\frac{k_L}{K}\to\omega_L,\, \frac{k_H}{K}\to\omega_H]{K,k_L,k_H\to\infty,} \int_{\omega_L}^{\omega_H} g\left(\mathcal{F}_\nu^{-1}(1-x)\right) dx = \int_{\mathcal{F}_\nu^{-1}(1-\omega_H)}^{\mathcal{F}_\nu^{-1}(1-\omega_L)} g(y)\, d\mathcal{F}_\nu(y)\ . \tag{A-5}$$

B Proof of Theorem 3.1

The following notation shall be used throughout this proof. Let the subset of indices of the J strongest users be denoted by

$$\mathcal{D}_J = \{i_1, \ldots, i_J\}. \tag{B-1}$$

The set of channel inputs due to the users in a subset $\mathcal{S}$ is denoted by $\overline{\mathcal{X}}_\mathcal{S}$. Considering in particular the subset of J strongest users, it follows that

$$\overline{\mathcal{X}}_{\mathcal{D}_J} = \{x_{i_1}, \ldots, x_{i_J}\}\ . \tag{B-2}$$

The set of channel inputs due to *all* received user *other* than those in the subset $\mathcal{S}$ is denoted by $\underline{\mathcal{X}}_\mathcal{S}$. In particular, it follows that

$$\underline{\mathcal{X}}_{\mathcal{D}_J} = \left\{x_{i_{J+1}}, \ldots, x_{i_K}\right\}\ . \tag{B-3}$$

The corresponding channel fading gains matrices and signatures matrices (see Section 2) are denoted, respectively, by

$$\overline{\mathbf{H}}_{\mathcal{D}_J} = \mathrm{diag}\left(h_{i_1}, \ldots, h_{i_J}\right) , \tag{B-4}$$

$$\overline{\mathbf{S}}_{\mathcal{D}_J} = \left[s_{i_1} \cdots s_{i_J}\right] , \tag{B-5}$$

$$\underline{\mathbf{H}}_{\mathcal{D}_J} = \mathrm{diag}\left(h_{i_{J+1}}, \ldots, h_{i_K}\right) , \tag{B-6}$$

$$\underline{\mathbf{S}}_{\mathcal{D}_J} = \left[s_{i_{J+1}} \cdots s_{i_K}\right] , \tag{B-7}$$

where s_{i_k} denotes the spreading sequence (signature) of the kth strongest user. For consistency, the set of channel outputs is denoted by $\mathcal{Y}$, and the total set of channel inputs is denoted by $\mathcal{X}^{\mathrm{T}}$. Finally, the (ordered) channel fading gains and signature matrices corresponding to *all* channel inputs are denoted by

$$\widetilde{\mathbf{H}} = \mathrm{diag}\left(h_{i_1}, \ldots, h_{i_K}\right) , \tag{B-8}$$

$$\widetilde{\mathbf{S}} = \left[s_{i_1} \cdots s_{i_K}\right] . \tag{B-9}$$

The first step in the proof is the derivation of the nonasymptotic capacity region. Towards this end, it is assumed that the spreading sequences are chosen randomly, and independently, *on a per symbol basis* (this setup differs for example from the case in which the signatures are randomly chosen *once* at the beginning of the transmission, and stay *unchanged*during the whole transmission period). The above assumption is introduced for technical reasons in order to simplify the derivations for the nonasysmptotic setup. It is emphasized however that this assumption has no effect on the final asymptotic result of Theorem 3.1, as both signature selection approaches yield the same results as the system size becomes asymptotically large.

In principle, the fact that some users are decoded, whereas the code structure of some users is not exploited, could lead to analytical difficulties. However, it is possible to sidestep those difficulties by using Kolmogorov's identity. The mutual information between the channel output and the information transmitted by the users to be decoded at the receiver

can be decomposed as

$$I(\overline{\mathcal{X}}_{\mathcal{D}_J};\mathcal{Y}) = I(\overline{\mathcal{X}}_{\mathcal{D}_J},\underline{\mathcal{X}}_{\mathcal{D}_J};\mathcal{Y}) - I(\underline{\mathcal{X}}_{\mathcal{D}_J};\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_J}) \tag{B-10}$$

$$= I(\mathcal{X}^{\mathrm{T}};\mathcal{Y}) - I(\underline{\mathcal{X}}_{\mathcal{D}_J};\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_J}) \tag{B-11}$$

$$= E_{\boldsymbol{S}}\left\{\log\det\left(\boldsymbol{I}+\bar{P}\widetilde{\mathbf{H}}^{\dagger}\widetilde{\mathbf{S}}^{\dagger}\widetilde{\mathbf{S}}\widetilde{\mathbf{H}}\right)\right\} - E_{\boldsymbol{S}}\left\{\log\det\left(\boldsymbol{I}+\bar{P}\underline{\mathbf{H}}^{\dagger}_{\mathcal{D}_J}\underline{\mathbf{S}}^{\dagger}_{\mathcal{D}_J}\underline{\mathbf{S}}_{\mathcal{D}_J}\underline{\mathbf{H}}_{\mathcal{D}_J}\right)\right\}, \tag{B-12}$$

where (B-12) follows from [17] and the independence of the information transmitted by different users, and $E_{\boldsymbol{S}}\{\cdot\}$ denotes expectation with respect to the realizations of the random signatures. It is emphasized that the fact that the signatures are known also for the users that are not decoded, is fully exploited in (B-11)–(B-12). Also, when spreading is used, unlike [10], it would be suboptimal to have a front end of J matched filters that neglects the $K-J$ weakest users.

Conditioned on the channel fades, the capacity region (bits/sec) determining the rates achievable by the J strongest users is given by:

$$C = \bigcap_{\mathcal{D}\subset\mathcal{D}_J}\left\{\sum_{i\in\mathcal{D}} R_i \le I(\overline{\mathcal{X}}_{\mathcal{D}_J};\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_J-\mathcal{D}})\right\}. \tag{B-13}$$

As can be seen from the following result, due to the particular structure of the considered problem, J out of the 2^J-1 rate constraint equations in (B-13) are sufficient in order to describe the capacity region.

Theorem B.1 *The capacity region achievable by the J strongest users is equal to (bits/sec)*

$$C = \bigcap_{j=1}^{J}\left\{\sum_{\ell=j}^{J} R_{i_\ell} \le I(\underline{\mathcal{X}}_{\mathcal{D}_{j-1}};\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_{j-1}}) - I(\underline{\mathcal{X}}_{\mathcal{D}_J};\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_J})\right\}. \tag{B-14}$$

Proof: First it is required to show that the following J equations are sufficient:

$$C = \bigcap_{j=1}^{J}\left\{\sum_{\ell=j}^{J} R_{i_\ell} \le I(x_{i_j},\ldots,x_{i_J};\mathcal{Y}\,|\,x_{i_1},\ldots,x_{i_{j-1}})\right\}. \tag{B-15}$$

To that end, proceeding as in (B-11), note that all the mutual informations arising in

(B-13) take the form

$$I(\overline{\mathcal{X}}_{\mathcal{D}_J}\,;\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_J-\mathcal{D}}) = I(\overline{\mathcal{X}}_{\mathcal{D}}\,;\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_J-\mathcal{D}}) \tag{B-16}$$

$$= I(\overline{\mathcal{X}}_{\mathcal{D}},\underline{\mathcal{X}}_{\mathcal{D}_J}\,;\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_J-\mathcal{D}}) - I(\underline{\mathcal{X}}_{\mathcal{D}_J}\,;\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_J}) \tag{B-17}$$

$$= E_{\mathcal{S}}\left\{\log\det\left(\boldsymbol{I}+\bar{P}\underline{\mathbf{H}}^{\dagger}_{\mathcal{D}_J-\mathcal{D}}\underline{\mathbf{S}}^{\dagger}_{\mathcal{D}_J-\mathcal{D}}\underline{\mathbf{S}}_{\mathcal{D}_J-\mathcal{D}}\underline{\mathbf{H}}_{\mathcal{D}_J-\mathcal{D}}\right)\right\} \tag{B-18}$$

$$- I(\underline{\mathcal{X}}_{\mathcal{D}_J}\,;\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_J})\,.$$

The right hand side of (B-18) is always lower bounded by the choice of $\mathcal{D}$ corresponding to the ℓ-weakest users among the J strongest users, i.e., the users of indices $i_{J-\ell+1},\ldots,i_{J-1},i_J$. Thus, the corresponding equation dominates. Note that the expression $I(\underline{\mathcal{X}}_{\mathcal{D}_J}\,;\mathcal{Y}\,|\overline{\mathcal{X}}_{\mathcal{D}_J})$ in (B-12) is fixed for a fixed J, and thus it is not affected by the specific choice of $\mathcal{D}\subset\mathcal{D}_J$.

Finally, generalizing the decomposition in (B-11), every mutual information appearing in (B-15) can be written as:

$$\begin{aligned} I(x_{i_j},\ldots,x_{i_J}\,;\mathcal{Y}\,|\,x_{i_1},\ldots,x_{i_{j-1}}) &= I(x_{i_j},\ldots,x_{i_J},\underline{\mathcal{X}}_{D_J}\,;\mathcal{Y}\,|\,x_{i_1},\ldots,x_{i_{j-1}}) \\ &\quad - I(\underline{\mathcal{X}}_{D_J}\,;\mathcal{Y}\,|\,x_{i_1},\ldots,x_{i_{j-1}},x_{i_j},\ldots,x_{i_J}) \\ &= I(\underline{\mathcal{X}}_{\mathcal{D}_{j-1}}\,;\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_{j-1}}) - I(\underline{\mathcal{X}}_{D_J}\,;\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_J}) \end{aligned} \tag{B-19}$$

for $j=1,\ldots,J$. It is noted that both mutual informations in the right hand side of (B-19) can be evaluated using the $E_{\mathcal{S}}\{\log\det(\cdot)\}$ expressions as in (B-12), yielding

$$\begin{aligned} I(\underline{\mathcal{X}}_{\mathcal{D}_{j-1}}\,;\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_{j-1}}) - I(\underline{\mathcal{X}}_{D_J}\,;\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_J}) &= E_{\mathcal{S}}\left\{\log\det\left(\boldsymbol{I}+\bar{P}\underline{\mathbf{H}}^{\dagger}_{\mathcal{D}_{j-1}}\underline{\mathbf{S}}^{\dagger}_{\mathcal{D}_{j-1}}\underline{\mathbf{S}}_{\mathcal{D}_{j-1}}\underline{\mathbf{H}}_{\mathcal{D}_{j-1}}\right)\right\} \\ &\quad - E_{\mathcal{S}}\left\{\log\det\left(\boldsymbol{I}+\bar{P}\underline{\mathbf{H}}^{\dagger}_{\mathcal{D}_J}\underline{\mathbf{S}}^{\dagger}_{\mathcal{D}_J}\underline{\mathbf{S}}_{\mathcal{D}_J}\underline{\mathbf{H}}_{\mathcal{D}_J}\right)\right\}\,. \end{aligned} \tag{B-20}$$

■

Following the derivation of the capacity region of the J strongest users, in the non-asymptotic scenario, it is required to examine the behavior of the $E_{\mathcal{S}}\{\log\det(\cdot)\}$ expressions in (B-20) as the system size becomes large (i.e., $K,N\to\infty$, $\frac{K}{N}\to\beta<\infty$). In particular, it is of interest to examine the expression

$$E_{\mathcal{S}}\left\{\log\det\left(\boldsymbol{I}+\bar{P}\underline{\mathbf{H}}^{\dagger}_{\mathcal{D}_L}\underline{\mathbf{S}}^{\dagger}_{\mathcal{D}_L}\underline{\mathbf{S}}_{\mathcal{D}_L}\underline{\mathbf{H}}_{\mathcal{D}_L}\right)\right\}\,, \tag{B-21}$$

where $\frac{L}{K} \to \delta$ $(0 \leq \delta \leq 1)$ at the asymptotic scenario. Towards this end, we review here the following result of [2].

Lemma B.2 (Shamai-Verdú, [2] Theorem IV.1) *Let the empirical distribution of the received powers of all users converge a.s. as $K, N \to \infty$, $\frac{K}{N} \to \beta < \infty$,t o some non-random limit $\mathcal{H}(P)$. Then, the spectral efficiency of the optimum multiuser detector is given by*

$$\tilde{C}_{opt} = \tilde{C}_{ms} + \log \frac{1}{\eta_{ms}} + (\eta_{ms} - 1) \log e \ , \tag{B-22}$$

where η_{ms} is the limiting multiuser efficiency of the corresponding linear MMSE detector, as given by Eq. (3-10)*, and*

$$\tilde{C}_{ms} = \beta E_{\mathcal{H}} \left\{ \log(1 + \eta_{ms} P) \right\} \tag{B-23}$$

is the spectral efficiency of this detector.

Defining

$$\mathcal{C}_{\text{opt}}(L\,;\bar{P}) \triangleq \frac{1}{N} E_{\boldsymbol{S}} \left\{ \log\det\left(\boldsymbol{I} + \bar{P} \underline{\mathbf{H}}^{\dagger}_{\mathcal{D}_L} \underline{\mathbf{S}}^{\dagger}_{\mathcal{D}_L} \underline{\mathbf{S}}_{\mathcal{D}_L} \underline{\mathbf{H}}_{\mathcal{D}_L} \right) \right\} \ , \tag{B-24}$$

it follows from Lemma B.2 that

$$\begin{aligned}
\mathcal{C}_{\text{opt}}(\delta\,;\bar{P}) &\triangleq \lim_{N,K\to\infty, \frac{K}{N}\to\beta, \frac{L}{K}\to\delta} \mathcal{C}_{\text{opt}}(L\,;\bar{P}) \\
&= \lim_{N,K\to\infty, \frac{K}{N}\to\beta, \frac{L}{K}\to\delta} \frac{1}{N} E_{\boldsymbol{S}} \left\{ \log\det\left(\boldsymbol{I} + \bar{P} \underline{\mathbf{H}}^{\dagger}_{\mathcal{D}_L} \underline{\mathbf{S}}^{\dagger}_{\mathcal{D}_L} \underline{\mathbf{S}}_{\mathcal{D}_L} \underline{\mathbf{H}}_{\mathcal{D}_L} \right) \right\} \\
&= \lim_{N,K\to\infty, \frac{K}{N}\to\beta, \frac{L}{K}\to\delta} \frac{1}{N} \sum_{m=L+1}^{K} \log\left(1 + \bar{P}\nu_{i_m} \eta_{\text{opt}}(L\,;\bar{P})\right) \\
&\qquad - \log \eta_{\text{opt}}(L\,;\bar{P}) + \left(\eta_{\text{opt}}(L\,;\bar{P}) - 1\right) \log e \ ,
\end{aligned} \tag{B-25}$$

where $\eta_{\text{opt}}(L\,;\bar{P})$ is the unique solution to the equation

$$1 = \eta + \frac{1}{N} \sum_{m=L+1}^{K} \frac{\bar{P}\nu_{i_m}\eta}{1 + \bar{P}\nu_{i_m}\eta} \ . \tag{B-26}$$

Using (A-5) and some algebra, it can be shown that

$$\lim_{N,K\to\infty,\frac{K}{N}\to\beta,\frac{L}{K}\to\delta} \eta_{\text{opt}}(L\,;\bar{P}) = \eta_{\text{opt}}(\delta\,;\bar{P})\ , \tag{B-27}$$

where $\eta_{\text{opt}}(\delta\,;\bar{P})$ is the unique solution to the equation

$$1-\beta(1-\delta) = \eta - \beta\Theta_{\text{opt}}(\delta\,;\bar{P}\eta)\ , \tag{B-28}$$

and

$$\begin{aligned}\Theta_{\text{opt}}(\omega\,;\varsigma) &= \int_{\mathcal{F}_\nu^{-1}(0)}^{\mathcal{F}_\nu^{-1}(1-\omega)} \frac{1}{1+\varsigma\phi}\, d\mathcal{F}_\nu(\phi) \\ &\underset{\text{Rayleigh}}{=} \int_0^{-\ln\omega} \frac{1}{1+\varsigma\phi}\, e^{-\phi}\, d\phi = \frac{1}{\varsigma}\, e^{\frac{1}{\varsigma}} \left[\mathcal{E}_1\!\left(\frac{1}{\varsigma}\right) - \mathcal{E}_1\!\left(\frac{1}{\varsigma} - \ln\omega\right)\right].\end{aligned} \tag{B-29}$$

The sum expression in (B-25) can be handled in an analogous manner, and defining

$$\begin{aligned}\Gamma_{\text{opt}}(w\,;\varsigma) &= \int_{\mathcal{F}_\nu^{-1}(0)}^{\mathcal{F}_\nu^{-1}(1-\omega)} \log\left(1+\varsigma\phi\right)\, d\mathcal{F}_\nu(\phi) \underset{\text{Rayleigh}}{=} \int_0^{-\ln\omega} \log\left(1+\varsigma\phi\right) e^{-\phi}\, d\phi \\ &= -\omega\log(1-\varsigma\ln\omega) + e^{\frac{1}{\varsigma}} \left[\mathcal{E}_1\!\left(\frac{1}{\varsigma}\right) - \mathcal{E}_1\!\left(\frac{1}{\varsigma} - \ln\omega\right)\right]\end{aligned} \tag{B-30}$$

it follows that

$$\mathcal{C}_{\text{opt}}(\delta\,;\bar{P}) = \beta\Gamma_{\text{opt}}\left(\delta\,;\bar{P}\eta_{\text{opt}}(\delta\,;\bar{P})\right) - \log\eta_{\text{opt}}(\delta\,;\bar{P}) + (\eta_{\text{opt}}(\delta\,;\bar{P}) - 1)\log e\ . \tag{B-31}$$

The final step in the proof of Theorem 3.1 is to plug the above asymptotic expressions in the capacity region (B-14). Recalling the underlying equal-rates assumption and (B-20), it follows that the non-asymptotic capacity region (*bits/sec/Hz*) becomes

$$\begin{aligned}\tilde{C} &= \bigcap_{j=1}^{J} \left\{(J-j+1)\tilde{R} \le \frac{1}{N} I(\underline{\mathcal{X}}_{\mathcal{D}_{j-1}}\,;\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_{j-1}}) - \frac{1}{N} I(\underline{\mathcal{X}}_{\mathcal{D}_J}\,;\mathcal{Y}\,|\,\overline{\mathcal{X}}_{\mathcal{D}_J})\right\} \\ &= \bigcap_{j=1}^{J} \left\{(J-j+1)\tilde{R} \le \mathcal{C}_{\text{opt}}(j-1\,;\bar{P}) - \mathcal{C}_{\text{opt}}(J\,;\bar{P}))\right\}\ .\end{aligned} \tag{B-32}$$

Letting $1-\mathcal{Q}$ and $1-x$ take the asymptotic role of $\frac{J}{K}$ and $\frac{j}{J}$, respectively, and taking the limits in (B-32), the outage constrained capacity ($J\tilde{R}$), for an FUU of $\mathcal{Q}$, is given by

$$\tilde{R}_{T_{\text{opt}}}(\mathcal{Q}) = \inf_{0<x\leq 1} \frac{\mathcal{C}_{\text{opt}}\left[(1-\mathcal{Q})(1-x)\,;\bar{P}\right] - \mathcal{C}_{\text{opt}}\left[(1-\mathcal{Q})\,;\bar{P}\right]}{x}\,, \tag{B-33}$$

where $\mathcal{C}_{\text{opt}}$ is given by (B-31). This completes the proof of Theorem 3.1. ■

C On the Minimum Receive $\frac{E_b}{N_0}$ Required for Reliable Communication

Following (3-3), the minimum required receive $\frac{E_b}{N_0}$ for reliable communication is given by [2]

$$\frac{E_b}{N_{0\,\min}} = \lim_{\bar{P}\to 0} \frac{\beta\bar{P}}{\tilde{R}_T(\beta,\bar{P})}\,, \tag{C-1}$$

where the notation $\tilde{R}_T(\beta,\bar{P})$ was introduced to emphasize that the outage constrained capacity is a function of $\bar{P}$ (the SNR) and the system load β. Starting with the linear detectors, it follows from (3-2) that

$$\tilde{R}_T \underset{\bar{P}\to 0}{=} \beta\bar{P}(1-\mathcal{Q})\mathcal{F}_\nu^{-1}(\mathcal{Q})\eta\log e + o(\bar{P})\,, \tag{C-2}$$

yielding

$$\frac{E_b}{N_{0\,\min}} = \frac{\ln 2}{(1-\mathcal{Q})\mathcal{F}_\nu^{-1}(\mathcal{Q})\eta}\,. \tag{C-3}$$

Observing (3-8) and (3-10), it follows that for both linear detectors $\lim_{\bar{P}\to 0}\eta = 1$, and hence

$$\frac{E_b}{N_{0\,\min}} = \frac{\ln 2}{(1-\mathcal{Q})\mathcal{F}_\nu^{-1}(\mathcal{Q})} \tag{C-4}$$

thus establishing (3-4).

Showing that the same result holds also for the "optimum detector" is less trivial, and the derivation follows the proof of Theorem 3.1 as given in Appendix B. Again the focus is on the asymptotic setup in which

$$K,N,L\to\infty,\ \frac{K}{N}\to\beta,\ \frac{L}{K}\to\delta\,, \tag{C-5}$$

however empirical averages and nonasymptotic notation shall be used whenever necessary for purposes of the derivation (the validity of this approach follows form the a.s. convergence of the limiting empirical distribution of the fading power levels, as stated in Section 2, in view of the underlying assumption of this report). Considering $\mathcal{C}_{\text{opt}}(L\,;\bar{P})$, as defined in (B-24) it follows from (B-25) and (B-26) that

$$\mathcal{C}_{\text{opt}}(L\,;\bar{P}) \triangleq \frac{1}{N}\sum_{m=L+1}^{K}\log\left(1+\bar{P}\nu_{i_m}\eta_{\text{opt}}(L\,;\bar{P})\right)-\log\eta_{\text{opt}}(L\,;\bar{P})+\left(\eta_{\text{opt}}(L\,;\bar{P})-1\right)\log e\ , \tag{C-6}$$

where $\eta_{\text{opt}}(L\,;\bar{P})$ is the unique solution to the equation

$$1=\eta+\frac{1}{N}\sum_{m=L+1}^{K}\frac{\bar{P}\nu_{i_m}\eta}{1+\bar{P}\nu_{i_m}\eta}\ . \tag{C-7}$$

As can be observed, the constant $\eta_{\text{opt}}(L\,;\bar{P})$ of satisfies

$$\eta \underset{\bar{P}\to 0}{=} \frac{1}{1+\bar{P}\,\frac{1}{N}\sum_{m=L+1}^{K}\nu_{i_m}}+o(\bar{P})\ . \tag{C-8}$$

Similarly, it follows that

$$\frac{1}{N}\sum_{m=L+1}^{K}\log\left(1+\bar{P}\nu_{i_m}\eta_{\text{opt}}(L\,;\bar{P})\right)\underset{\bar{P}\to 0}{=}\bar{P}\eta_{\text{opt}}(L\,;\bar{P})\log e\frac{1}{N}\sum_{m=L+1}^{K}\nu_{i_m}+o(\bar{P})\ . \tag{C-9}$$

Substituting the above two limiting expressions in (C-6) yields

$$\mathcal{C}_{\text{opt}}(L\,;\bar{P})\underset{\bar{P}\to 0}{=}\beta\bar{P}\frac{1}{K}\sum_{m=L+1}^{K}\nu_{i_m}+o(\bar{P})\ , \tag{C-10}$$

where the notation $\frac{K}{N}=\beta$ was used for simplicity (and in agreement with the underlying assumptions regarding the asymptotic scenario).

Now as shown in Appendix B (see (B-32)), the capacity region of the "optimum detector" is given by

$$C=\bigcap_{j=1}^{J}\left\{(J-j+1)\tilde{R}\le\mathcal{C}_{\text{opt}}(j-1\,;\bar{P})-\mathcal{C}_{\text{opt}}(J\,;\bar{P})\right\}\ . \tag{C-11}$$

Observing (C-10), the dominantin equality in (C-11) is clearly the one for which $j = J$, and the maximum attainable rate per user (under outage constraint) must satisfy

$$\tilde{R} = \mathcal{C}_{\text{opt}}(J-1\,;\bar{P}) - \mathcal{C}_{\text{opt}}(J\,;\bar{P})\ . \tag{C-12}$$

Hence, recalling that $\tilde{R}_T = J\tilde{R}$, the outage constrained capacity of the "optimum detector" satisfies

$$\tilde{R}_{T_{\text{opt}}} \underset{\bar{P}\to 0}{=} \beta\bar{P}\frac{J}{K}\nu_{i_J}\log e + o(\bar{P})\ , \tag{C-13}$$

which in the asymptotic setup of (C-5) (recalling also that $\frac{J}{K} \to 1-\mathcal{Q}$ in this case) satisfies

$$\tilde{R}_{T_{\text{opt}}} \underset{\bar{P}\to 0}{=} (1-\mathcal{Q})\mathcal{F}_\nu^{-1}(\mathcal{Q})\beta\bar{P}\log e + o(\bar{P})\ . \tag{C-14}$$

(C-4) then follows immediately from (C-1).

D Proof of Theorem 4.2

The proof of Theorem 4.2 follows a similar path to that of the proof of Theorem III.4 in [2]. The underlying idea in the proof is that one can tell whether the optimum β is finite or not for a given $\frac{E_b}{N_0}$, by examining the behavior of the outage constrained capacity as $\beta \to \infty$, while $\bar{P}$ is adjusted to comply with the given $\frac{E_b}{N_0}$. To that end, as in [2], we fix an arbitrary $d \triangleq \beta\bar{P} \geq 0$. As discussed in Section 4, the outage constrained capacity of the linear MMSE detector coincides as $\beta \to \infty$ with the spectral efficiency of the matched filter detector, and satisfies

$$\lim_{\beta\to\infty} \tilde{R}_{T_{\text{ms}}}(\mathcal{Q}) = (1-\mathcal{Q})\mathcal{F}_\nu^{-1}(\mathcal{Q})\log e - \left(\frac{E_b}{N_0}\right)^{-1}\ . \tag{D-1}$$

Therefore, using the fact that (see (3-3))

$$d = \beta\bar{P} = \tilde{R}_{T_{\text{ms}}}(\mathcal{Q})\frac{E_b}{N_0}\ , \tag{D-2}$$

it is immediately observed that (fixing $\mathcal{Q}$) the constant d uniquely determines $\frac{E_b}{N_0}$, at the limit as $\beta \to \infty$, through the relation

$$d = (1-\mathcal{Q})\mathcal{F}_\nu^{-1}(\mathcal{Q}) \log e \frac{E_b}{N_0} - 1 \ . \tag{D-3}$$

The next step is to examine the derivative of the outage constrained capacity (3-2) with respect to $\frac{1}{\beta}$, or equivalently, with respect to $\bar{P}$, and find the value of d for which the derivative becomes 0 at $\bar{P} = 0$. The rationale behind the above investigation is as follows. For a fixed $\frac{E_b}{N_0} > \frac{E_b}{N_{0\,\mathrm{cr}}}$ the outage constrained capacity attains a maximum at some finite value of β, while for $\frac{E_b}{N_0} < \frac{E_b}{N_{0\,\mathrm{cr}}}$ the outage constrained capacity monotonically grows with β. Therefore, relying on the continuity of the outage constrained capacity with respect to β, investigating its derivative, as discussed above, reveals the critical $\frac{E_b}{N_0}$ at which the maximum is *attained* for $\frac{1}{\beta} = 0$, which implies $\beta \to \infty$. With that in mind, the aim is to find the value of d for which the following holds:

$$\frac{\partial}{\partial \bar{P}} \left[\frac{d}{\bar{P}} (1-\mathcal{Q}) \log \left(1 + \mathcal{F}_\nu^{-1}(\mathcal{Q}) \eta(\bar{P}) \bar{P} \right) \right] \Bigg|_{\bar{P}=0} = 0 \ , \tag{D-4}$$

where the notation $\eta(\bar{P})$ is used to emphasize the dependency of the multiuser efficiency in $\bar{P}$. Now following some algebra, it can be shown that for (D-4) to hold it is required that

$$\dot{\eta}(\bar{P})|_{\bar{P}=0} = \frac{1}{2} \mathcal{F}_\nu^{-1}(\mathcal{Q}) \eta^2(\bar{P})|_{\bar{P}=0} \ , \tag{D-5}$$

where $\dot{\eta}$ denotes the derivative of η with respect to $\bar{P}$. In order to proceed from here we invoke the following Lemma.

Lemma D.1 ([2], Property 9) *Fix $d = \beta\bar{P} \geq 0$, then the behavior of the multiuser efficiency of the linear MMSE detector for $\bar{P} \to 0$ is*

$$\eta(\bar{P}) = \frac{1}{1+d} + \frac{d\, E_{\mathcal{F}_\nu}\{\nu^2\}}{(1+d^3)} \bar{P} + o(\bar{P}) \ . \tag{D-6}$$

Using (D-6) in (D-5) and solving for d yields

$$d = \frac{\mathcal{F}_\nu^{-1}(\mathcal{Q})}{2E_{\mathcal{F}_\nu}\{\nu^2\} - \mathcal{F}_\nu^{-1}(\mathcal{Q})} \ . \tag{D-7}$$

Finally, substituting (D-7) in (D-3) it follows that

$$\frac{E_b}{N_0} = \frac{2E_{\mathcal{F}_\nu}\{\nu^2\}}{2E_{\mathcal{F}_\nu}\{\nu^2\} - \mathcal{F}_\nu^{-1}(\mathcal{Q})} \frac{E_b}{N_{0\,\min}} \underset{\text{Rayleigh}}{=} \frac{4}{4 + \ln(1-\mathcal{Q})} \frac{E_b}{N_{0\,\min}}, \tag{D-8}$$

where $\frac{E_b}{N_{0\,\min}}$ is given by (3-4).

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